

A PRELIMINARY STUDY ON THE INHIBITORY ACTION OF MYSTATIN ON THE PROMASTIGOTE FORMS OF *LEISHMANIA TROPICA* AND *LEISHMANIA INFANTUM* ISOLATES OF PAKISTAN

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ABSTRACT

Leishmania is the etiological agent of a group of parasitic disease Leishmaniasis. Recent reports show that the disease is on the rise in Pakistan. As no vaccine is yet available for any type of leishmaniasis, chemotherapy is the only means of controlling the disease. As a part of our project to find out new antileishmanial compounds, we have studied the antileishmanial activity of Nystatin (mycostatin), a polyene compound, on the promastigote forms of *L. tropica* and *L. infantum* isolates of Pakistan. In *in vitro* cultivation, the EC50 values (the concentration of drug necessary to inhibit the growth rate of cells to 50% of the control value) obtained for Nystatin against the promastigote of *L. tropica* and *L. infantum* were 19.5-19.8 iu/ml. The mode of action proposed for this drug is same as for famous leishmanicidal drug Amphotericin B as both of these polyene drugs strongly and selectively binds to the sterols present on the surface of the parasite result in the formation of unusual pores/gaps on the surface that results in the leakage of the ions. This leakage finally leads to the destruction of the cell.

INTRODUCTION

The parasitic protozoan *Leishmania* is the causative agent of a group of parasitic disease i.e. Leishmaniasis (Yasinzai *et al.*, 1996; Baily, 1994; Sendino, 1993). The parasites are transmitted by different species of blood sucking sandflies of the genus phlebotomus (Mutinga, 1996). The disease originally discovered at the turn of this century, has not been eradicated; rather, it is on the rise in many parts of the world (Magill, 1995). Leishmaniasis is generally classified in three major types. The cutaneous leishmaniasis generally caused by *L. major/tropica* species of the parasite; the mucocutaneous leishmaniasis which is caused by *L. braziliensis* species of the parasite and visceral leishmaniasis or more commonly known as "kala azar" which is caused by *L. donovani* species. Cutaneous, mucocutaneous and macrophages of spleen, liver and bone marrow are infected by the parasite in above mentioned three

types of leishmaniasis respectively (Yasinzai *et al.*, 1966; Baily, 1994). The parasite has a digenic life cycle; flagellated, motile promastigote present in female sandfly vector or in *in vitro* culture and aflagellated, intracellular amastigote present in macrophages of the host (Yasinzai, 1996).

The pentavalent antimonial drugs introduced in the early days of the chemotherapy of leishmaniasis are still the drugs of choice (Koff and Rosen, 1994). However the recent clinical report show an alarming trend. A considerable proportion of cases (15-25%) are becoming unresponsive to antimonial therapy (Giri, 1994). In such cases, pentamidine, biguanidine and amphotericin-B serve as second line of defence (Giri, 1994). All of these drugs however are highly toxic and needs monitored conditions that can only be provided in an advanced hospital situation (Koff and Rosen, 1994; Giri, 1994). Due to these limitation, researchers are looking forward for new antileishmanial drugs that are effective, cheap and above all exhibit least toxic effects.

MATERIAL AND METHODS

Parasites [*L. major* (MOHM/PK/91/ABDULLA) and *L. infantum* (MHOM/PK/91/RAB-KARAMATVL)] were isolated from the infected patients and maintained unmodified Tobie's medium between 21-23°C in dark. Parasites were then sub-cultured in the medium 199 (Sigma) supplemented with 10% heat inactivated foetal bovine serum (HIFBS) and 2% sterile human urine between 21-23°C in an cooled incubator for 72 hours for bulk cultivation.

The parasites were counted under 40X magnification on a modified neubaur chamber and were sedimented down at 3000g for 15 minutes. The sedimented parasites were diluted to a concentration of 2×10^6 parasites per ml with fresh medium and were transferred aseptically into sterile 5 ml capacity culture tubes.

Nystatin suspension was obtained from Lederle/Cyanamid Pakistan. Each ml of Nystatin suspension contained 100,000 iu (international units) of the active compound. The suspension was aseptically introduced into the culture tube and was serially diluted so that different tubes contained 10000, 5000, 2500, 1250, 625, 312.5, 156.15, 78, 39, 19.5, 9.75 and 0 iu of the active compound. The tubes were placed in an incubator between 21-22°C for upto 72 hrs during which control organisms multiply 3-6 times. The parasites were counted on modified neubaur chamber and inhibition was assessed.

RESULTS

All tests were conducted in triplicate and the mean of three readings was con-

sidered as the final result. Standard error was also calculated and has been shown in the graphs. Results are graphically represented in Fig.1. for *L. tropica* and *L. infantum* respectively. It was found that mainly 19.5 iu per ml was enough to inhibit the growth of 50% of the parasites of *L. tropica* while EC50 value found against *L. infantum* was found more or less the same.

DISCUSSION

Mucocutaneous form of the disease is fortunately not present however cutaneous and visceral forms are prevalent in many parts of the country. Recent reports show that the disease is on the rise (Yasinzai, 1996; Magill, 1995). As no vaccine is yet available for any type of leishmaniasis, chemotherapy is the only means of controlling the disease. Although natural products have not been extensively screened for their leishmanicidal activity however trends are appeared to be changed (Iwu, *et al*; Olliaro and Bryceson, 1993). Many novel compounds isolated from various medicinal plants have been reported for their leishmanicidal activity in last few years (Iwu, *et al.*, 1994; Chan *et al.*, 1993; Schlein *et al.*, 1994; Fournet *et al.*, 1994; Kapil, 1994; Munoz *et al.*, 1994; al-Jitvai *et al.*, 1995).

The cytoplasm of all living things is bounded by the cytoplasmic membrane, which serves as selective permeability barrier, carries out active transport functions, and thus controls the internal composition of the cell. If the functional integrity of the cell membrane is disrupted, macromolecule and ions escape from the cell, and cell damages or death ensues. As microbial cell surface is different than the animal cell surface, it is possible to disrupt it by certain agents. Consequently, selective chemotherapy is possible (Berman, 1985).

Nystatin or more commonly known as Mycostatin is a polyene drug (New *et al.*, 1981; Sessa and Weishman, 1967). This is produced by the growth of *Streptomyces noursei*. Nystatin-A is closely related to the Amphotericin-B, another polyene antifungal drug. Like Amphotericin-B, the Nystatin is a macrocyclic lactone with a ketal ring, an all-trans polyene system and a mycoseamine (3- amino-3-deoxyrhamnose) moiety. Nystatin is completely ineffective against bacteria, however it is quite active against various type of fungus and yeast (Hammond, 1977; Sessa and Weishman, 1967). The mode of action proposed for this drug is similar to that of Amphotericin-B (Hammond, 1977; Sessa and Weishman, 1967).

Amphotericin-B and Nystatin specifically interacts with ergosterol in fungal membranes (Hammond, 1977), *Leishmania* sterols are comparable to fungal sterols, and efficacy of this drug against *Leishmania* may similarly be due to interaction with ergosterol in *Leishmania* membrane (Nes and McKean, 1977). *Leishmania* synthesize

sterols. The major sterol synthesized by the parasite is 5-dehydroepisterol, episterol, ergosterol and some other sterols (Pieragostini *et al.*, 1974; Grecken *et al.* (1976). Like Amphotericin-B, the Nystatin strongly and selectively binds to the sterols present on the surface of the parasite particularly the ergosterol and results in the formation of unusual pores/gaps on the surface of the parasite that results in the leakage of the ions. This leakage eventually leads to the destruction of the cell. At the same time it is also suggested that while on the surface of the parasite, the Nystatin and Amphotericin-B also inhibit the further synthesis of the sterols (Hammond, 1977). The Amphotericin-B is the second line of the drug used against the parasites when initial treatment of the antimonial drugs get fail, however, the toxic effects (Hammond, 1977; Sessa, 1967) of this drug like fever, chills, nausea, vomiting, renal failure, hypokalemia, and anaemia are quite prominent as well. Nystatin is of course not as active as the Amphotericin-B but is far less toxic and much more suitable for the oral administration.

As antileishmanial drug, Nystatin is at its very initial stages, however the strong leishmanicidal activity of the compound on promastigote of two different species *in vitro* was observed. Whether such inhibition occurs in amastigotes and accounts for antileishmanial effect of this compound *in vivo* remains to be determined. Hopefully such investigations would lead to more specific antileishmanial agents and to a major improvement in our anti leishmanial armamentarium.

Fig. 1: Inhibitory action of Nystatin on the promastigote forms of *L. tropica* and *L. infantum*.

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